

AI Textbook Q&A System

A RAG-Based Educational Question Answering System
with Deep Source Tracing & ROS 2 Voice Integration

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CST8507: Natural Language Processing
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Part 1: RAG Pipeline

Part 2: ROS 2 Voice

1. Introduction & Motivation

Problem > Goal > Our Solution

PROBLEM

Students face 46+ AI/ML textbooks
Hard to navigate across books
No unified search system
No way to verify answer sources

GOAL

Ask natural language questions
Get accurate, sourced answers
Deep source tracing:
 Book > Chapter > Page > Region
Click to see original PDF

SOLUTION

RAG pipeline with 4 retrievers
RRF fusion for ranking
Ollama LLM (qwen2.5:0.5b)
React UI + PDF viewer
ROS 2 voice integration

Research Question: How can we build a transparent, traceable educational Q&A system for AI/ML learning?

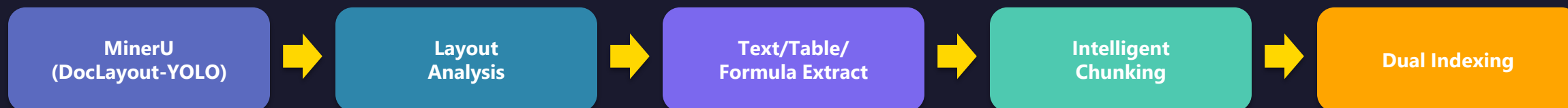
2. Dataset - 46 AI/ML Textbooks

Curated collection of canonical educational resources



46 BOOKS | 85,356 CHUNKS | ~500MB PDFs | Open-access / educational copies

Preprocessing Pipeline



2. Dataset - Library View

Browse all 46 textbooks
with cover art, author, and page count

AI Textbook Q&A
RAG PIPELINE

SearchLibrary

Ddia

Eloquent Javascript

Fastapi Modern Web

Fluent Python

Grl

Http Definitive Guide

Mathematics for Machine Le...

MI Fundamentals

Nlp

Pattern Recognition and Ma...

Practical Binary Analysis

Pro Git

Probabilistic Machine Learni...

Python Cookbook

Python Testing Pytest

Release It

Serious Cryptography

Snell Probability

Speech and Language Proce...

46
BOOKS

85,356
CHUNKS

Library
46 textbooks in collection

Search by title, author, or keyword...



Practical Binary Analysis

Dennis Andriess

2018 · 460 pages



Serious Cryptography

Jean-Philippe Aumasson

2017 · 314 pages



Bayesian Reasoning and Machine Learning

David Barber

2012 · 610 pages



SSH, The Definitive Guide

Daniel Barrett et al.

2005 · 560 pages



TypeScript Deep Dive

Basarat Ali Syed

2020 · 145 pages



Python Cookbook

Beazley

134 pages



Pattern Recognition and Machine Learning

Christopher M. Bishop

2006 · 758 pages



Convex Optimization

Boyd

714 pages



Pro Git

Chacon

501 pages



CLRS

Cormen

1312 pages



Mathematics for Machine Learning

Marc Peter Deisenroth et al.

2020 · 417 pages



Think Python 2e

Downey

244 pages



Think Stats 2e

Downey

264 pages



Nlp

Eisenstein

587 pages



Web Scalability

Ejsmont

417 pages



Art Of Postgresql

Fontaine

457 pages



Sre

Google

550 pages



Swe

Google

602 pages



Http Definitive Guide

Gourley

658 pages



Snell Probability

Grinstead

518 pages



Grl

Hamilton

141 pages



The Elements of Statistical Learning

Trevor Hastie et al.

2009 · 764 pages



Eloquent Javascript

Haverbeke

435 pages



Speech and Language Processing

Dan Jurafsky & James H. Ma...

2024 · 626 pages



MI Fundamentals

Kelleher

853 pages



Ddia

Kleppmann

613 pages



Using Sqlite

Kreibich

528 pages



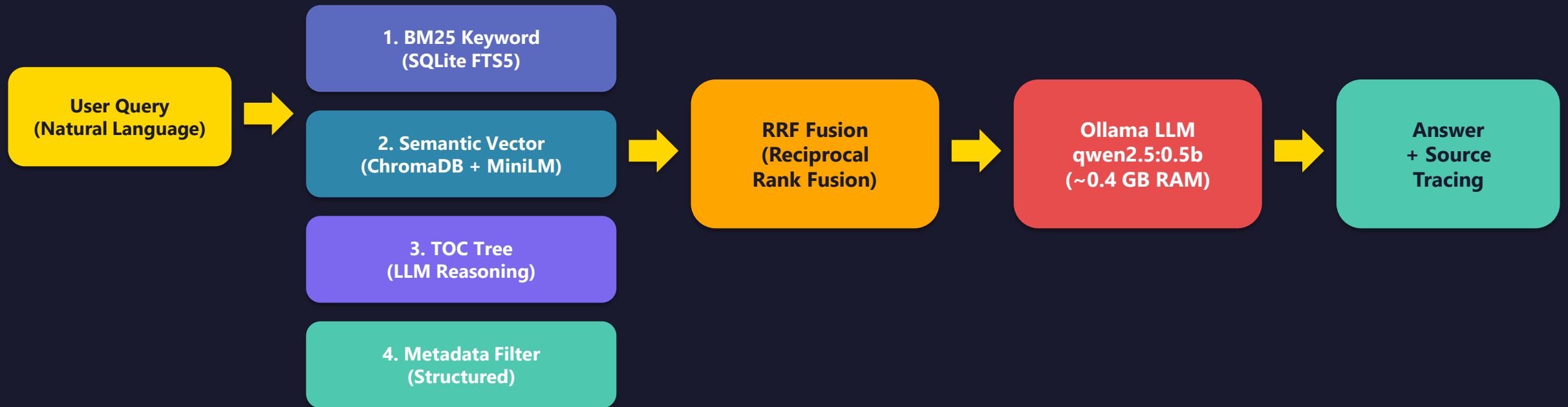
Fastapi Modern Web

Lubanovic

280 pages

3. RAG Architecture

4-Method Hybrid Retrieval + Reciprocal Rank Fusion



BM25: Fast exact keyword matching via SQLite FTS5 extension

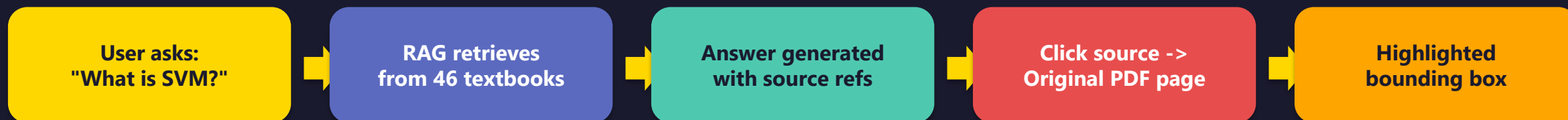
Semantic: Embedding-based retrieval with all-MiniLM-L6-v2 + ChromaDB

TOC Tree: Hierarchical TOC navigation using LLM reasoning per book

Metadata: Structured filtering by book, chapter, page, content type

4. Key Feature - Deep Source Tracing

Not just citations - pixel-level traceability to original documents



What makes this different from basic RAG?

Standard RAG: "According to textbook X, SVM is..."

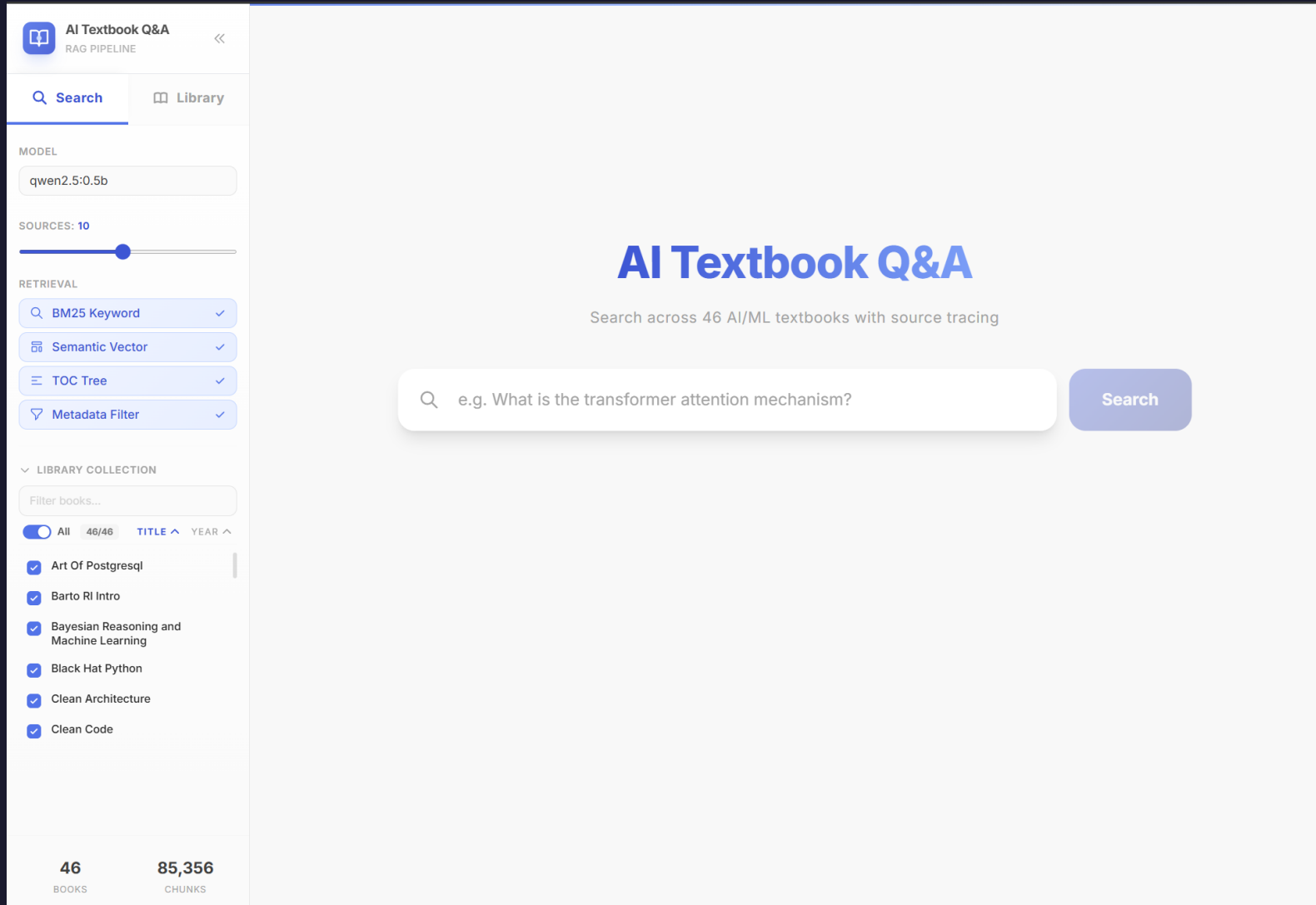
Our system: Click "View PDF (p.354)" -> opens original page with yellow highlight

Each chunk stores: book_title, chapter, page_number, bounding_box (x, y, w, h)

MinerU (DocLayout-YOLO) extracts spatial coordinates during preprocessing

Enables instant visual verification against original source material

5. React UI - Search Interface



5. React UI - Results & PDF Viewer

AI Textbook Q&A
RAG PIPELINE

Search Library

MODEL
qwen2.5:0.5b

SOURCES: 10

RETRIEVAL
BM25 Keyword ✓
Semantic Vector ✓
TOC Tree ✓
Metadata Filter ✓

LIBRARY COLLECTION
Filter books...
All 46/46 TITLE YEAR
Art Of Postgresql
Barto Rl Intro
Bayesian Reasoning and Machine Learning
Black Hat Python
Clean Architecture
Clean Code

46 BOOKS
85,356 CHUNKS

Ask a Question

Search across 46 AI/ML textbooks with source tracing

what is SVM?

Search

RELATED
Kernel Methods
Logistic Regression
Hyperplane
Margin
Regularization

ANSWER
SVM stands for Support Vector Machine. It is a supervised learning algorithm used to classify or distinguish between different classes of data points. The SVM model can be trained using kernel methods such as linear and quadratic kernels, which allow it to handle non-linearly separable data sets by finding the optimal hyperplane that separates the classes.

SOURCE BOOKS (10)
Pattern Recognition and Machine Learning
Christopher M. Bishop · 1 match · Best score: 0.7531
Page 354 · 7.1.1 Overlapping class distri... fusion Score: 0.7531
where $\mathcal{M}$ denotes the set of indices of data points having $0 < a_n < C$. An alternative, equivalent formulation of the support vector machine, known as the ν -SVM, has been proposed by Schölkopf *et al.* (2000). This involves maximizing
View PDF (p.354)
Probabilistic Machine Learning: An Introduction
Kevin P. Murphy · 1 match · Best score: 0.5493
Bayesian Reasoning and Machine Learning
David Barber · 1 match · Best score: 0.5000
Understanding Machine Learning
Shai Shalev-Shwartz & Shai Ben-David · 1 match · Best score: 0.3280
The Elements of Statistical Learning

Pattern Recognition and Machine Learning

354 / 758 1.0x

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334 7. SPARSE KERNEL MACHINES

model (7.13). The remaining data points constitute the support vectors. These have $a_n > 0$ and hence from (7.25) must satisfy

$$t_n y(\mathbf{x}_n) = 1 - \xi_n. \quad (7.35)$$

If $a_n < C$, then (7.31) implies that $\mu_n > 0$, which from (7.28) requires $\xi_n = 0$ and hence such points lie on the margin. Points with $a_n = C$ can lie inside the margin and can either be correctly classified if $\xi_n \leq 1$ or misclassified if $\xi_n > 1$.

To determine the parameter b in (7.1), we note that those support vectors for which $0 < a_n < C$ have $\xi_n = 0$ so that $t_n y(\mathbf{x}_n) = 1$ and hence will satisfy

$$t_n \left(\sum_{m \in \mathcal{S}} a_m t_m k(\mathbf{x}_n, \mathbf{x}_m) + b \right) = 1. \quad (7.36)$$

Again, a numerically stable solution is obtained by averaging to give

$$b = \frac{1}{N_{\mathcal{M}}} \sum_{n \in \mathcal{M}} \left(t_n - \sum_{m \in \mathcal{S}} a_m t_m k(\mathbf{x}_n, \mathbf{x}_m) \right) \quad (7.37)$$

where $\mathcal{M}$ denotes the set of indices of data points having $0 < a_n < C$.

An alternative, equivalent formulation of the support vector machine, known as the ν -SVM, has been proposed by Schölkopf *et al.* (2000). This involves maximizing

$$\tilde{L}(\mathbf{a}) = -\frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N a_n a_m t_n t_m k(\mathbf{x}_n, \mathbf{x}_m) \quad (7.38)$$

subject to the constraints

$$0 \leq a_n \leq 1/N \quad (7.39)$$

$$\sum_{n=1}^N a_n t_n = 0 \quad (7.40)$$

$$\sum_{n=1}^N a_n \geq \nu. \quad (7.41)$$

This approach has the advantage that the parameter ν , which replaces C , can be interpreted as both an upper bound on the fraction of *margin errors* (points for which $\xi_n > 0$ and hence which lie on the wrong side of the margin boundary and which may or may not be misclassified) and a lower bound on the fraction of support vectors. An example of the ν -SVM applied to a synthetic data set is shown in Figure 7.4. Here Gaussian kernels of the form $\exp(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2)$ have been used, with $\gamma = 0.45$.

Although predictions for new inputs are made using only the support vectors, the training phase (i.e., the determination of the parameters $\mathbf{a}$ and b) makes use of the whole data set, and so it is important to have efficient algorithms for solving

5. React UI - Feature Details

All features of the interactive search interface

Search Features

- 4 retrieval methods with individual toggles
- Adjustable top-K sources (1-20 slider)
- Library collection filter (select books)
- Model selector (qwen2.5:0.5b)
- Related topic tag suggestions
- Answer with full-text generation
- Source documents with relevance scores

PDF Viewer Features

- Click "View PDF" -> opens inline viewer
- Page navigation (prev/next, page input)
- Zoom controls (fit width, zoom in/out)
- One-page / two-page spread mode
- Highlighted source region (yellow bbox)
- Full-screen viewing mode
- Scroll to navigate through pages

Tech: React (Vite) + FastAPI backend + SQLite FTS5 + ChromaDB + Ollama + MuPDF

6. Evaluation - 20 Test Questions

Manual assessment across AI/ML/NLP domains

Scoring Method

1.0 = Correct answer with relevant sources

0.5 = Partially correct / incomplete

0.0 = Incorrect or irrelevant

Each question: top 3 retrieved documents

Relevance marked for each source chunk

Final accuracy = mean score across 20 Qs

20 Test Questions

1. What is neural attention?
2. Generative vs discriminative?
3. Gradient for logistic regression?
4. What is an N-gram LM?
5. What is skip-gram?
6. RNNs as language models?
7. 3D rigid body transformation?
8. Weight initialization?
9. Network depth?
10. Cost-sensitive classification?
11. What is TF-IDF?
12. What is backpropagation?
13. Transformer architecture?
14. What is Word2Vec?
15. Named Entity Recognition?
16. What is beam search?
17. Dropout regularization?
18. Batch normalization?
19. What is transfer learning?
20. What is BLEU score?

Result: **69.6%** avg accuracy (target >80%) | Domains: ML, NLP, CV, Math, Deep Learning

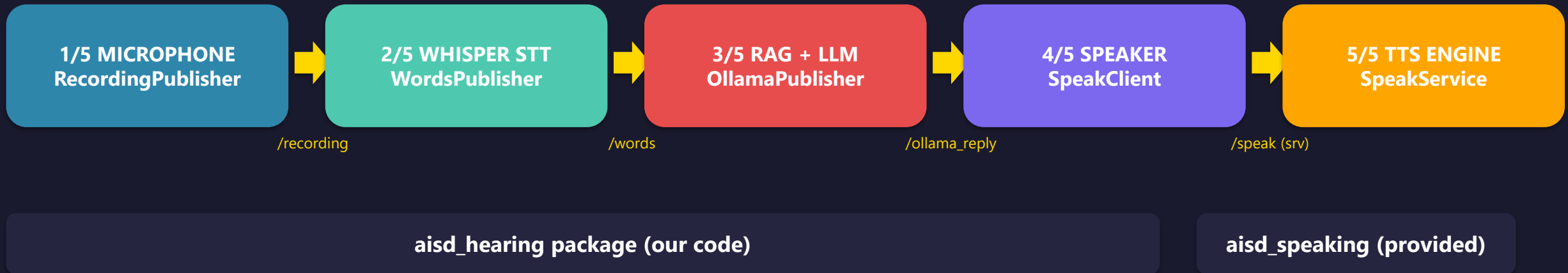
Part 2

ROS 2 Voice Integration

Converting the RAG pipeline into a voice-interactive system
using ROS 2 Humble on Ubuntu 22.04

7. ROS 2 Architecture - 5-Node Pipeline

Voice input -> STT -> RAG -> TTS -> Voice output



Node Details:

1. RecordingPublisher: Mic input + Silero VAD
2. WordsPublisher: faster-whisper (int8, auto-lang)
3. OllamaPublisher: RAG + qwen2.5:0.5b (our node)
4. SpeakClient: Receives reply, calls TTS service
5. SpeakService: gTTS -> audio playback

Deployment:

Platform: Ubuntu 22.04 (loaner laptop)
ROS 2 Humble + colcon build
Process mgmt: tmux (5 panes)
Deploy: SCP from Windows -> laptop
One-click: start_loaner.bat

8. Implementation Details

ollama_publisher.py - The bridge between Part 1 and Part 2

ollama_publisher.py (our new node):

Subscribes to: /words (STT output)

Publishes to: /ollama_reply (LLM answer)

Loads knowledge.txt at startup

Constructs system prompt with RAG context

Queries local Ollama API (localhost:11434)

15s cooldown between replies

Hallucination filtering on input

Deployment Automation:

start_loaner.bat (Windows -> laptop):

Step 1: SCP sync (4 .py + knowledge.txt)

Step 2: Stop existing tmux session

Step 3: colcon build --symlink-install

Step 4: tmux launch 5 panes

stop_loaner.bat:

Kill tmux + cleanup orphan processes

9. ROS 2 Demo - Live Voice Q&A

5-node pipeline running on loaner laptop via tmux

Q&A flow (attention, skip-gram questions) → Continued responses + TTS playback confirmation

Pipeline: Mic → Whisper → RAG + Ollama → gTTS → Speaker

```
Windows PowerShell
ound detected, level: 11378)
[INFO] [1774889418.077818783] [recording_publisher]: [Recording] Stopped: Time limit reached (15s)
[INFO] [1774889418.265702785] [recording_publisher]: [Recording] Finished: 237449 samples captured
[INFO] [1774889418.293709649] [recording_publisher]: [File] Saved: 1774889403014371304.wav
[INFO] [1774889418.294547723] [recording_publisher]: [Recording] Completed: 463.8 KB
[INFO] [1774889418.295476604] [recording_publisher]: [Publish] File published to ROS topic: 1774889403014371304.wav
[INFO] [1774889418.296356202] [recording_publisher]: [Recording] Starting new recording...
[INFO] [1774889418.306161975] [recording_publisher]: [Recording] Waiting for sound input... (speak now)
[INFO] [1774889418.695053777] [recording_publisher]: [Recording] Started (audio level: 7263)

probability distribution over input positions."
[INFO] [1774889389.468102076] [ollama_publisher]: [Words] Received: "What is skip gram in world 2 bag? Yes, the key component in an n-gram language model is the scoring function that calculates attention. Wait using a 2-layer B-forward level and then a..."
[INFO] [1774889389.468835612] [ollama_publisher]: [Ollama] Sending query to local Ollama (qwen2.5:0.5b)...
[INFO] [1774889396.603342265] [ollama_publisher]: [Ollama] Response generated (276 chars)
[INFO] [1774889396.604080373] [ollama_publisher]: [Reply] Published: "Skip-gram is a word embedding method used in Word2Vec to predict surrounding context words given a target word. It uses a two-layer feedforward network with a B (batch size) layer as the encoder and outputs it through an affine transformation with W (weights) as hidden state."
[INFO] [1774889399.686648436] [ollama_publisher]: [Cooldown] Ignoring input (3s < 15s since last reply)
[INFO] [1774889411.494232905] [ollama_publisher]: [Cooldown] Ignoring input (15s < 15s since last reply)

o produce a probability distribution over input positions. This mechanism enables the model to focus on relevant parts of the input when making predictions.": "OK"
[INFO] [1774889335.791559436] [speak_client]: [TTS] Response for "The model uses attention to focus on specific elements of the input data without being overwhelmed by all the details.": "OK"
[INFO] [1774889336.567145599] [speak_client]: [Words] Received: "The key components of an N-gram language model are the scoring function that computes attention weights through a two-layer feedforward network, followed by softmax activation for outputting probabilities."
[INFO] [1774889352.288429263] [speak_client]: [TTS] Response for "The key components of an N-gram language model are the scoring function that computes attention weights through a two-layer feedforward network, followed by softmax activation for outputting probabilities.": "OK"
[INFO] [1774889367.202729827] [speak_client]: [Words] Received: "Yes, the key component in an N-gram language model is the scoring function that calculates attention weights using a two-layer feedforward network and then applying softmax activation to produce a probability distribution over input positions."
[INFO] [1774889386.836035691] [speak_client]: [TTS] Response for "Yes, the key component in an N-gram language model is the scoring function that calculates attention weights using a two-layer feedforward network and then applying softmax activation to produce a probability distribution over input positions.": "OK"
[INFO] [1774889396.606467327] [speak_client]: [Words] Received: "Skip-gram is a word embedding method used in Word2Vec to predict surrounding context words given a target word. It uses a two-layer feedforward network with a B (batch size) layer as the encoder and outputs it through an affine transformation with W (weights) as hidden state."
[INFO] [1774889420.768099281] [speak_client]: [TTS] Response for "Skip-gram is a word embedding method used in Word2Vec to predict surrounding context words given a target word. It uses a two-layer feedforward network with a B (batch size) layer as the encoder and outputs it through an affine transformation with W (weights) as hidden state.": "OK"

[hlp_rag] 0:nodes*
"aisd" 16:50 30-Mar-26
```

Results - Key Achievements

Hybrid RAG

Successfully implemented hybrid RAG architecture with 4 retrieval methods + RRF fusion

Deep Source Tracing

Deep source tracing with clickable PDF references and highlighted bounding box regions

Layout-Aware

Layout-aware document processing preserving tables, formulas, and figures

ROS 2 Integration

Full ROS 2 integration with vocal interaction pipeline (5-node architecture)

Low Memory

Low memory footprint (<1.5GB total) suitable for robot deployment

10. Challenges & Solutions

Whisper Hallucinations

Silero VAD pre-filtering + custom hallucination detector (repetition check, known phrases, trigram analysis)

Echo Loop (TTS to Mic)

15-second cooldown timer after each reply prevents system from hearing its own TTS output

Resource Constraints

qwen2.5:0.5b (0.4GB) + faster-whisper int8 -- 3 AI models (Whisper + Ollama + gTTS) on one CPU laptop

Cross-Platform Deploy

Automated bat/sh scripts: SCP sync -> colcon build -> tmux launch -- one-click deployment from Windows to Ubuntu laptop

PDF Layout Parsing

MinerU + DocLayout-YOLO detects 10 element categories -- preserves tables, formulas, figures with bounding box coordinates

11. Discussion & Future Work

Current Limitations

Small LLM (0.5B params) - limited generation quality; accuracy 69.6% vs 80% target due to model size constraints

Single-turn Q&A only (no conversation memory)

Static knowledge file (no live updates)

CPU inference is slow (~10-15s per response)

ROS 2 Part 2 uses simplified knowledge base

English-only knowledge base content

Future Work

Upgrade to qwen2.5:3b with GPU support

Multi-turn conversation with context memory

Dynamic knowledge base updates via API

Multi-language support (KR, CN, EN)

Integration with iRobot Create 3 hardware

Fine-tuning on educational domain data

Live Demo

AI Textbook Q&A
RAG PIPELINE

SearchLibrary

MODEL
qwen2.5:0.5b

SOURCES: 11

RETRIEVAL
BM25 Keyword
Semantic Vector
TOC Tree
Metadata Filter

LIBRARY COLLECTION
Filter books...
All 46/46 TITLE ^ YEAR v
TypeScript Deep Dive
Practical Binary Analysis
Serious Cryptography
Bayesian Reasoning and Machine Learning
SSH, The Definitive Guide
Python Cookbook

46 BOOKS
85,356 CHUNKS

AI Textbook Q&A

Search across 46 AI/ML textbooks with source tracing

Conclusion

We built a RAG-based educational Q&A system that:

- Answers AI/ML questions using 46 canonical textbooks

- Provides deep source tracing to exact PDF pages with highlighted regions

- Implements 4-method hybrid retrieval with RRF fusion

- Integrates seamlessly with ROS 2 for voice-interactive robot deployment

- Runs locally with low memory footprint (<1.5GB)

A complete voice-interactive RAG pipeline - from 46 textbooks to real-time voice Q&A on a ROS 2 platform.

Thank You!

Questions?

GitHub: github.com/LannieYoo/nlp_project

Python | Ollama | ChromaDB | SQLite | React | FastAPI | ROS 2 | MinerU | Whisper | gTTS